

ECON 101—Introduction to Microeconomics
Fall 2019

Final Exam

November 27, 2019

Part I

Please record your answers here. Each question is worth 3 points.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20

Part II

Please record your answers here. Each question is worth 4 points.

1	2	3	4	5	6	7	8	9	10

MCQ 1 (60)	MCQ 2 (40)	Total (100)
For instructor's use only. Do not write anything in here.		

Part I

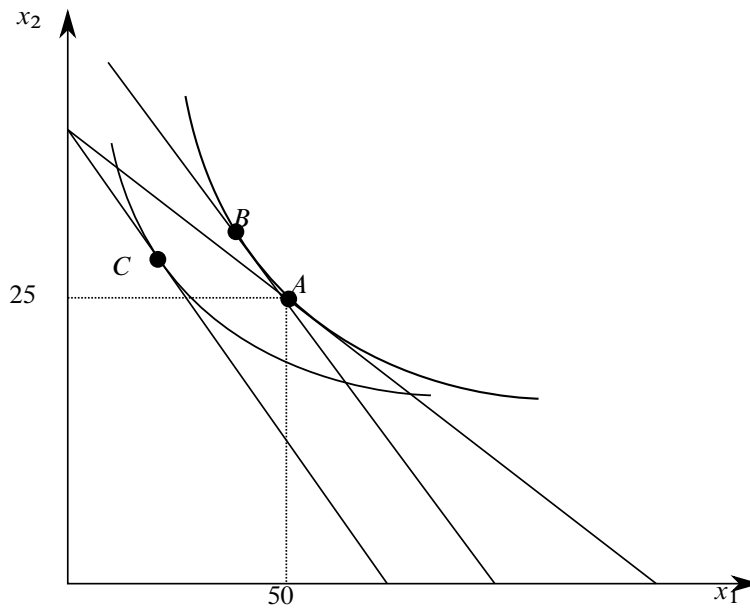
Please record all questions in the cover sheet. Each question is worth 3 points for a total of 60 points.

Version	◇	♡	♠	♣	\$
Question					
1	(d)	(d)	(a)	(b)	(a)
2	(c)	(c)	(b)	(d)	(c)
3	(c)	(d)	(b)	(a)	(c)
4	(b)	(b)	(b)	(b)	(a)
5	(d)	(c)	(d)	(a)	(d)
6	(c)	(d)	(c)	(c)	(a)
7	(c)	(d)	(b)	(c)	(a)
8	(c)	(b)	(d)	(a)	(a)
9	(b)	(c)	(a)	(a)	(b)
10	(b)	(b)	(d)	(a)	(d)
11	(b)	(d)	(b)	(d)	(b)
12	(a)	(b)	(d)	(a)	(a)
13	(d)	(a)	(d)	(c)	(d)
14	(b)	(c)	(a)	(a)	(b)
15	(b)	(d)	(c)	(b)	(d)
16	(a)	(a)	(c)	(b)	(b)
17	(b)	(b)	(d)	(b)	(a)
18	(d)	(c)	(a)	(a)	(d)
19	(a)	(d)	(d)	(b)	(c)
20	(b)	(d)	(a)	(c)	(a)

Part II

Please record all questions in the cover sheet. Each question is worth 4 points for a total of 40 points.

Consider the following scenario as depicted by the following figure.



This consumer can consume goods x_1 and x_2 . The consumer has an income of \$100 dollars. The price of x_1 is \$1 dollar and the price of x_2 is \$2. Each consumer initially chooses the initial consumption bundle $x_1 = 50, x_2 = 25$ which is denoted by point A. Suppose that there is an increase in the price of x_1 to \$4. After this price increase, the optimal consumption bundle of the consumer is denoted by point C. Suppose the government gives the consumer a monetary transfer of \$80 so that she can achieve her original indifference curve before the price change. Also, assume that an economic researcher in the bureau of taxation gives you the following information:

- The arc own price elasticity for good 1 of the consumer is $|\epsilon_{x_1}^{p_1}| = 1.11$.
- The arc income elasticity with respect to good 1 of the consumer is $\epsilon_{x_1}^I = 1.5$

1. The substitution effect is:

- (a) -30 (b) -50 (c) -25 (d) -15

2. The income effect is:

- (a) -20 (b) -10 (c) -30 (d) -15

3. The arc cross-price elasticity of good x_2 with respect to price p_1 is:

- (a) -0.30 (b) 0.15 (c) 1 (d) 0.2

In some states of the United States of America, legislation has changed, giving the now the same treatment to marijuana and alcohol. The consumption, production and exchange of marijuana with recreational purposes is now legal.

Suppose that initially the demand and supply of marijuana were the following

$$Q_D = 1000 - 2P$$

$$Q_P = -100 + 2P$$

With the new legislation the production costs of marijuana have decreased and the demand has also increased so that the equations of demand and supply after the approval of the new legislation are:

$$Q_D = 1200 - 2P$$

$$Q_S = 2P$$

With the previous information answer the following three questions:

4. If the objective of the state governments is that the same quantity of marijuana is consumed before and after the legislation comes into force, the per unit tax on marijuana needs to be _____, a _____% of the tax will need to be paid by producers, and _____% of the tax will need to be paid by consumers.
 - (a) \$150; 50%; 50%.
 - (b) \$150; 75%; 25%.
 - (c) \$300; 50%; 50%.
 - (d) \$300; 75%; 25%.
5. After the drug legalization, which of the following quantities maximizes government revenues? The percentage of the tax that will be paid by producers is _____% and the the percentage paid by consumers is _____%.
 - (a) 225; 50%; 50%.
 - (b) 225; 75%; 25%.
 - (c) 300; 50%; 50%.
 - (d) 300; 75%; 25%.
6. Which of the following options minimizes the dead-weight loss associated with the imposition of a tax on marijuana?
 - A tax of \$200 per unit.
 - A tax of \$300 per unit.
 - A tax of \$150 per unit.
 - A tax of \$100 per unit.

Suppose that there are two firms, each with a total cost function $TC(q) = 10q$ and facing a demand function of $P = 130 - 2Q$, where Q is total output. The marginal cost of the firm is of course $MC(q) = 10$.

7. Suppose that the firms are operating as a cartel, dividing evenly profits and the aggregate level of production. The monopoly price is _____, each firm produces _____, and each firms earns profits of _____.
 - (a) \$10;30;\$0
 - (b) \$70;15;\$900
 - (c) \$60;17.5;\$876
 - (d) \$30;25;\$500

8. Now suppose that the firms have two possible strategies: to produce half of the cartel aggregate quantity or to produce 20 units. If both firms produce 20 units, the market price is _____ and each firms earns _____ in profits.
 - (a) \$30;\$500
 - (b) \$70, \$900
 - (c) \$50; \$800
 - (d) \$60; \$876

9. If one firm is producing half the cartel aggregate quantity and the other is producing 20 units, the firm producing half the cartel quantity has profits of _____, and the firm producing 20 units has profits of _____.
 - (a) \$876;\$900
 - (b) \$500; \$600
 - (c) \$200; \$1200
 - (d) \$750; \$1000

10. The Nash equilibrium of the game is _____.
 - (a) For each firm to produce 20 units.
 - (b) For each firm to produce 15 units.
 - (c) For each firm to produce 30 units.
 - (d) Fr each firm to produce 10 units.